FINM3013 Introduction to Financial Derivatives
Solution to Quiz 1

I. Multiple choice questions (10 points):

1. A
2. D
3. A
4. C. $(1.155 – 1.15) \times 125{,}000 = 625$.
5. C. To increase the beta by 0.3 we need to long $0.3 \times 36{,}000{,}000/(900 \times 250)$ or 48 contracts.

II. Problems (30 points)

1. (12 points)

The profit table is

	$S_T < 30$	$S_T > 30$
Long 1 share	$S_T – 30$	$S_T – 30$
Long 1 put option	$30 – S_T – 5$	–5
Total	–5	$S_T – 35$

a) -5(1000) = 5000 loss
b) -5(1000) = 5000 loss
c) (35-35)1000 = 0
d) (37-35)1000 = 2000 gain

2. (8 points)

There is a margin call if $1000 is lost on the contract. This will happen if the price of wheat futures falls by 20 cents from 200 cents to 180 cents per bushel. $1500 can be withdrawn if the futures price rises by 30 cents to 230 cents per bushel.

3. (10 points)

Yes, there is an arbitrage opportunity. We enter into a long forward contract and short sell one share, investing 70% of the proceeds at 8% and paying the remaining 30% as a security deposit to attract interest at 4%. At the time of delivery the cash investments plus interest will be worth 17*0.7*1.08 + 17*0.3*1.04 = 18.156, out of which $18 will need to be paid for one share to close out the short position in stock. This leaves a $0.156 arbitrage profit.

The rate *d* for the security deposit such that there is no arbitrage opportunity satisfy 30%*17* (1 + d) + 70%*17*1.08 $\leqslant$ 18. The highest such rate is d = 0.941%.